

DERIVATIVES RESEARCH REPORT

The Derivatives Time Machine

Could a Gulf airline's 2023 hedge have protected it from the 2026 oil shock?

A no-hindsight audit of Brent futures, Black-76 options, and jet-fuel basis risk

Result: PARTIALLY CORRECT

Illustrative derivatives education project

Executive finding

VERDICT

Partially correct

The hedge protected the 2026 shock and reduced cost variance, but it did not save money over the entire 35-month audit.

The short answer. A risk manager who approved the July 2023 hedge would have been pleased in March 2026 and disappointed over the full window. The locked hybrid saved **\$7.76 million** in March and **\$1.42 million** across January-June 2026. Yet it added **\$3.02 million** to cost across all 35 months.

Decision test	Threshold	Observed	Result
Full-window savings	> \$0	-\$3.02m	Fail
2026 shock savings	> \$0	+\$1.42m	Pass
Cost variance reduction	> 0%	7.5%	Pass
Peak-month protection	Material	+\$7.76m	Pass

Current recommendation

As of 27 July 2026: HEDGE. Cover 70% of next month's policy volume with 104 long Brent futures and 78 zero-cost collars. For each collar, buy the **\$96.72 call** and sell the **\$88.24 put**; leave 30% unhedged.

This is an illustrative procurement hedge, not personal investment advice. Live execution requires current exchange quotes, liquidity and margin checks, airline-specific volume forecasts, and governance approval.

1. Research design: the time-capsule test

The project separates prediction from hindsight. All hedge parameters are selected from information available through 31 July 2023. That configuration is frozen before the audit begins. Realized prices from August 2023 through June 2026 are then passed through the unchanged derivative payoff rules.

Layer	Information date	Purpose
Training	Jan 2018-Jul 2023	Estimate cross-hedge beta and choose policy-compliant weights
Locked recommendation	31 Jul 2023	Freeze hedge ratio, strikes, mix, costs, and decision rule
Out-of-sample audit	Aug 2023-Jun 2026	Measure realized costs with no parameter hindsight
Current refresh	27 Jul 2026	Re-estimate today's one-month trade ticket; not included in the audit

The hypothetical exposure

The airline consumes 10 million U.S. gallons of jet fuel each month. Physical cost is the EIA U.S. Gulf Coast kerosene-type jet-fuel spot price. Brent is a cross-hedge, so the model first converts Brent into dollars per gallon and estimates the sensitivity of jet-fuel changes to Brent changes.

LOCKED BETA

0.870x

Each 1.0 cent move in Brent's gallon-equivalent price was associated with an estimated 0.87 cent move in jet fuel during training.

Locked policy mix

Component	Share	Economic role
Unhedged	30%	Retains benefit if prices fall
Long Brent futures	40%	Linear protection against rising Brent
Long calls	0%	Optimizer did not select standalone calls inside the policy
Zero-cost collars	30%	Caps upside while selling some downside benefit

Governance constraints were set before testing: futures could not exceed 40%, option share had to be at least 30%, and at least 20% of volume had to remain unhedged. The objective minimized mean net fuel cost plus 75% of the excess 95% conditional tail cost.

2. The 35-month audit

The locked hybrid tracked the physical fuel bill closely for most of the audit, then diverged sharply during the 2026 spike and reversal. That is exactly the behavior a derivatives audit should expose: protection depends on path, strike placement, and basis—not simply on whether oil is eventually higher or lower.



Figure 1. Monthly net cost of the locked hybrid versus the unhedged jet-fuel spot cost.

Strategy	Total cost	Savings	CVaR95 / gal	Efficiency
Unhedged	\$856.3m	\$+0.0m	\$4.126	0.0%
Futures	\$861.9m	\$-5.7m	\$4.121	2.5%
Long call	\$858.3m	\$-2.1m	\$3.952	11.2%
Collar	\$858.8m	\$-2.5m	\$4.039	7.0%
Locked hybrid	\$859.3m	\$-3.0m	\$3.989	7.5%

The standalone call delivered the best variance reduction (11.2%) and the lowest 2026 shock-window cost, but option premium made it lose \$2.07 million over the full audit. The policy-constrained hybrid reduced variance by 7.5% and lowered 95% tail cost by \$0.137 per gallon, while losing \$3.02 million in cumulative cost.

3. What the 2026 shock revealed

The stress event made the project unusually informative. EIA reported an exceptionally sharp petroleum-price increase in the first quarter of 2026 and described continued Middle East shipping disruptions in the second quarter. The audit therefore tests the locked hedge against both a violent rally and a fast reversal.

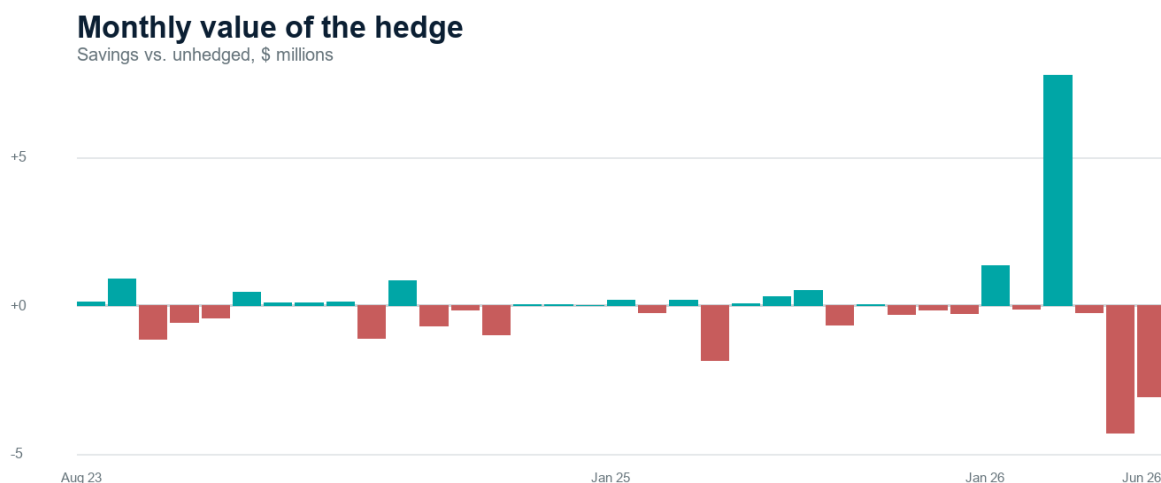


Figure 2. Positive bars are monthly savings; negative bars are the cost of hedging versus remaining unhedged.

Month	Unhedged / gal	Hybrid / gal	Hedge value
2026-01	\$2.277	\$2.142	\$+1.353m
2026-02	\$2.428	\$2.438	\$-0.103m
2026-03	\$3.965	\$3.189	\$+7.764m
2026-04	\$4.287	\$4.311	\$-0.244m
2026-05	\$3.237	\$3.666	\$-4.292m
2026-06	\$2.867	\$3.173	\$-3.059m

March: the thesis worked

Brent settled at \$126.69 per barrel and jet fuel at \$3.965 per gallon. The hedge cut that month's net cost to \$3.189 per gallon, protecting \$7.76 million on the assumed 10-million-gallon volume.

May-June: the trade-off arrived

Prices reversed quickly. Linear futures and the short-put side of the collar surrendered the benefit of cheaper physical fuel, costing \$4.29 million in May and \$3.06 million in June versus remaining unhedged.

April: basis risk mattered

Jet fuel reached \$4.287 per gallon while the hybrid cost was \$4.311. The March-entered call strike was above the subsequent Brent settlement, so the derivative did not fully offset the refined-product stress. This is the central modeling lesson: Brent is a strong but imperfect hedge for jet fuel.

4. Was the 2023 call correct?

JUDGMENT

Partially correct

The recommendation passed the insurance test and failed the full-window savings test.

What was right

Tail timing. The hedge delivered its largest benefit in the very month the physical fuel bill surged most sharply.

Risk reduction. Hybrid cost variance fell 7.5%, and conditional tail cost declined by \$0.137 per gallon.

Policy discipline. Keeping 30% unhedged prevented the hedge from becoming a disguised outright oil position.

What was wrong

Cumulative economics. The hedge cost \$3.02 million across 35 months; a claim that it 'saved money' would be false.

Reversal exposure. Futures and short puts were painful when prices fell after the shock.

Basis risk. Brent option payoffs did not always match jet-fuel crack-spread movements.

The most defensible conclusion

The 2023 decision should be defended as a risk-budget choice, not as a forecast of profitable trading P&L. If the board's objective was guaranteed cumulative savings, the hedge failed. If the objective was to buy resilience during a severe fuel-price shock, it worked—but at a measurable insurance cost.

5. Current one-month hedge ticket

Current input	Value	As of
Brent spot	\$91.82/bbl	2026-07-27
Jet-fuel spot	\$3.582/gal	2026-07-27
OVX	60.62%	2026-07-27
1M Treasury	3.80%	2026-07-27
Refreshed cross-hedge beta	1.095x	60-month window

HEDGE 70%

ACTION

The refreshed optimizer reproduces the policy mix: 40% futures, 30% zero-cost collar, 30% unhedged.

Leg	Action	Quantity	Strike / reference
Brent future	Buy / long	104 contracts	Synthetic 1M future \$92.12
Collar call	Buy	78 contracts	Strike \$96.72
Collar put	Sell	78 contracts	Strike \$88.24
Open volume	Remain unhedged	30% policy share	Benefits if fuel prices fall

Why hedge now

OVX was 60.62%, signaling expensive but material expected oil volatility. A pure long-call program would preserve downside but pay a large premium. The collar finances the call by giving up some benefit below the put strike, while the 30% open share keeps meaningful participation in falling prices.

Execution controls

Control	Required action
Basis	Compare live jet-fuel crack spread with Brent before sizing
Liquidity	Validate bid-ask depth at both option strikes and chosen expiry
Volume	Replace 10m-gallon teaching assumption with forecast lift and burn
Governance	Confirm counterparty, margin, VaR and hedge-accounting limits

6. Formula appendix and limitations

Core equations

Cross-hedge beta: $\beta = \text{Cov}(\Delta\text{Jet}, \Delta\text{Brent}/42) \div \text{Var}(\Delta\text{Brent}/42)$.

Synthetic future: $F_0 = S_0 \exp(rT)$.

Black-76 call: $C = \exp(-rT)[F_0 N(d_1) - KN(d_2)]$, with $d_2 = d_1 - \sigma\sqrt{T}$.

Consumer futures cost: Jet settlement – $\beta(\text{Brent settlement} - F_0)/42$ + friction.

Hybrid cost: 30% unhedged + 40% futures + 30% collar.

Important limitations

Limitation	Effect on interpretation
Synthetic futures	EIA spot prices settle a cost-of-carry future; the study does not claim actual historical exchange fills.
Volatility proxy	OVX reflects USO options, not a complete Brent volatility surface.
Constant volume	Monthly consumption is fixed at 10m gallons; real airline demand is seasonal and uncertain.
Simplified costs	Credit, margin, tax, accounting, liquidity and counterparty effects are excluded.
Monthly rebalancing	Intramonth cash-flow and margin stress are not modeled.
Current ticket	The July 2026 recommendation is prospective and has not yet been audited.

Reproducibility

The Python program fetches public inputs, estimates the hedge, recreates every audit period, and exports the scorecard. The Excel model reproduces the Black-76 pricing and payoff formulas in visible cells, includes a current Goal Seek input for the zero-cost put strike, and exposes model checks. The web dashboard shows only decision-ready results.

Sources

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Data cutoff and interpretation

The latest common market observation used for the current ticket is 27 July 2026. The completed audit ends on 30 June 2026 so that partial July observations do not contaminate performance. Figures may differ by rounding; the underlying Python and Excel calculations retain full precision.